

Instructions:

- **Due April 21, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW09_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework develops students' ability to analyze feedback control systems within the basic servo-loop framework using transfer-function models. Students will construct and interpret root-locus, Bode, and Nyquist plots to evaluate closed-loop stability, robustness, and stability margins. They will determine ranges of proportional gains that guarantee asymptotic stability and relate frequency-domain characteristics to closed-loop pole behavior. The assignment also strengthens proficiency in using MATLAB tools to support and verify analytical reasoning.

Problem 1. Consider the systems

$$G_1(s) = 10, \quad (1)$$

$$G_2(s) = s + 1, \quad (2)$$

$$G_3(s) = \frac{1}{s + 10}. \quad (3)$$

1. For each transfer function, sketch the Bode magnitude and phase plots using straight-line asymptotic approximations. Identify the low-frequency and high-frequency asymptotes and mark the break (corner) frequencies where applicable.

[Hint: The break frequency is the frequency at which the low- and high-frequency asymptotes intersect.]

2. Add the asymptotic magnitude and phase contributions from the three systems obtained in part (a). This provides an approximation of the Bode plot of the combined system

$$G(s) = G_1(s)G_2(s)G_3(s). \quad (4)$$

3. Using the MATLAB command `bode`, plot the Bode magnitude and phase plots of each individual system and of the combined system

$$G(s) = G_1(s)G_2(s)G_3(s). \quad (5)$$

Verify that the analytical approximation obtained in the previous part agrees with the MATLAB Bode plot.

Problem 2. Consider the unstable system

$$G(s) = \frac{1}{(s - 1)(s + 5)(s + 10)}. \quad (6)$$

Suppose a proportional controller $G_c(s) = K_p = 60$ is used in the basic servo loop (BSL). The loop transfer function is therefore

$$L(s) = K_p G(s). \quad (7)$$

1. Draw the root locus of $L(s)$ using Matlab. Is the BSL asymptotically stable? Explain why? If the BSL is asymptotically stable, calculate the gain margin using the root locus plot.
2. Draw the Bode plot of $L(s)$ using Matlab. Is the BSL asymptotically stable? Explain why? If the BSL is asymptotically stable, calculate the gain margin and the phase margin using the Bode plot. [**Hint:** Use the matlab function `margin`.]
3. Draw the Nyquist plot of $L(s)$ using Matlab. Is the BSL asymptotically stable? Explain why? If the BSL is asymptotically stable, calculate the gain margin using the Nyquist plot.

Hint:

- (a) Use the Matlab function `[re,im,wout] = nyquist(sys,w)` to evaluate the transfer function at several frequencies.
- (b) Gain margin is $1/g$, where g is the point at which the Nyquist plot and the negative real axis intersect.
- (c) To find the phase margin, calculate the magnitude of `rm + j im` for each frequency `wout`. Select the complex number for which the magnitude is one, and compute its angle. Let this angle be denoted by ϕ_1 . Phase margin is $\phi_m = 180 + \phi_1$.

Problem 3. For the following loop transfer functions

$$1. L(s) = \frac{1}{s-2},$$

$$2. L(s) = \frac{1}{s+2},$$

$$3. L(s) = \frac{s-1.5}{(s+5.5)(s-2)(s-3.5)},$$

use the Nyquist plot to determine whether the closed-loop transfer function is asymptotically stable. If the closed-loop transfer function is asymptotically stable, determine the smallest and the largest value of proportional gain that yields an asymptotically stable closed-loop. If the closed-loop transfer function is not asymptotically stable, determine the value of the proportional gain that yields an asymptotically stable closed-loop. [**Hint:** Use `nyquist(G,frequency range)` to plot the Nyquist plot for the desired frequency range.]